MINI PROJECT DOCUMENTATION

AI HACKATHON TEAM MATCHER

AI-Based Team Formation Platform

Project Type	Web Application
Domain	Artificial Intelligence / Web Development / Education Technology
Development Platform	Base44 AI
Core Technology	AI-Based Matching
Primary Users	B.Tech / College Students
Repository	match-hack-syn

> “Build the right team. Match the right skills. Win together.”

Prepared as a B.Tech Mini Project Documentation

AI HACKATHON TEAM MATCHER

AI-Based Team Formation Platform

Project Description: An AI-powered web application that matches students based on their skills, interests, preferred roles, experience, and complementary abilities to form balanced and effective hackathon teams.

Development Platform: Base44 AI

Project Type: Web Application

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1. ABSTRACT

Hackathons provide students with an opportunity to solve real-world problems through technology. However, one of the major challenges faced by students is finding suitable teammates with complementary skills.

The AI Hackathon Team Matcher is a web-based application designed to solve this problem using Artificial Intelligence. The system analyzes student profiles, including technical skills, interests, preferred roles, experience levels, and soft skills. Based on this information, the application recommends suitable teammates and generates compatibility scores.

Unlike traditional team formation systems that mainly match users based on similar interests, this application focuses on complementary abilities. For example, an AI/ML student can be matched with a frontend developer, backend developer, UI/UX designer, or DevOps student to create a well-balanced team.

The application was developed using Base44 AI, which helps create the web application interface, workflows, database structure, authentication, and AI-powered functionality.

The goal of this project is to make hackathon team formation faster, smarter, and more effective.

2. INTRODUCTION

Hackathons are becoming increasingly popular among college students. Successful hackathon projects usually require multiple skills such as:

- Programming
- Artificial Intelligence
- Machine Learning
- Web Development
- UI/UX Design
- Database Management
- Cloud Computing
- Communication
- Presentation

A single student may not possess all these skills. Therefore, forming a team with complementary skills is important.

Currently, students generally find teammates through friends, college groups, WhatsApp groups, social media, or personal networks. These methods may not always produce balanced teams.

The proposed AI Hackathon Team Matcher provides an intelligent platform where students can create profiles and receive AI-based teammate recommendations.

3. PROBLEM STATEMENT

Problem

Students participating in hackathons often struggle to find teammates who have the right combination of technical and non-technical skills.

Existing informal methods of finding teammates:

- Depend heavily on personal contacts
- May result in teams with duplicate skills
- Do not analyze skill gaps
- Do not consider complementary abilities
- Do not provide compatibility scores
- Do not provide intelligent team recommendations

Proposed Solution

Develop an AI-powered web application that:

1. Collects student skill and interest information.
2. Analyzes individual student profiles.
3. Identifies complementary skills.
4. Calculates team compatibility.
5. Recommends suitable teammates.
6. Generates balanced team structures.
7. Identifies missing skills within a team.
8. Provides AI-based explanations for recommendations.

4. OBJECTIVES

The main objectives of the project are:

Primary Objectives

- To develop an AI-based hackathon team formation platform.
- To match students based on complementary skills.
- To improve the quality of hackathon teams.
- To reduce the time required to find suitable teammates.
- To provide explainable AI-based recommendations.

Secondary Objectives

- Identify skill gaps within teams.

- Recommend suitable roles to students.
- Analyze project requirements.
- Recommend additional members when required.
- Provide team compatibility scores.
- Create a centralized platform for student collaboration.

5. SCOPE OF THE PROJECT

The system can be used by:

- B.Tech students
- CSE students
- IT students
- College coding clubs
- Hackathon participants
- Innovation clubs
- Student project teams
- Hackathon organizers

The application can later be extended to:

- College-level hackathons
- National hackathons
- Internship team formation
- Project team formation
- Startup team formation
- Research collaboration
- Skill-based professional networking

6. EXISTING SYSTEM

In the existing system, students generally find hackathon teammates through:

- Friends
- College groups
- WhatsApp groups
- Telegram groups
- Social media
- Personal contacts
- Coding communities

Limitations of Existing System

1. No intelligent matching.

2. No skill compatibility analysis.

3. Difficult to identify missing skills.

4. Students may form teams with overlapping skills.

5. Limited personalization.

6. No compatibility score.

7. No AI explanation for team recommendations.

7. PROPOSED SYSTEM

The proposed system uses AI-based matching to recommend suitable teammates.

Basic Workflow

Student Registration ↓ Create Student Profile ↓ Add Skills & Interests ↓ Select Preferred Role ↓ AI Profile Analysis ↓ Skill Complementarity Analysis ↓ Compatibility Score ↓ Recommended Teammates ↓ Team Formation ↓ Team Skill Analysis ↓ Project Analysis ↓ AI Team Intelligence Report

8. KEY FEATURES

8.1 Student Registration

Students can create an account using:

• Name

• Email

• College

• Branch

• Year

• Profile information

8.2 Student Profile

Students can create detailed profiles containing:

Technical Skills

• Python

• Java

• C/C++

• JavaScript

• React

• Node.js

• SQL

• Machine Learning

• Deep Learning

• NLP

- Generative AI
- Cloud
- DevOps
- Cybersecurity
- UI/UX

Soft Skills

- Leadership
- Communication
- Teamwork
- Creativity
- Problem Solving
- Presentation
- Time Management

Interests

- Artificial Intelligence
- Healthcare
- Education
- FinTech
- Cybersecurity
- Agriculture
- Sustainability
- Space Technology
- Smart Cities

9. AI TEAM MATCHING

This is the main feature of the project.

The system analyzes multiple factors:

- Technical skills
- Complementary skills
- Interests
- Preferred roles
- Experience
- Soft skills
- Availability
- Hackathon domain

The system then generates a compatibility score.

Example

Student A:

Python + Machine Learning + NLP

Student B:

React + UI/UX

Student C:

Node.js + SQL

Student D:

AWS + Docker + DevOps

The AI can identify these students as a strong team because their skills complement each other.

10. COMPATIBILITY SCORE

The application provides a team compatibility score from 0–100%.

Example:

Team Compatibility: 92%

Score Breakdown

Factor	Score
Skill Complementarity	95%
Interest Compatibility	90%
Role Compatibility	94%
Experience Balance	88%
Soft Skills	92%
Availability	91%

The score helps students understand why a particular teammate is recommended.

11. EXPLAINABLE AI

A major feature of the application is explainable recommendations.

Instead of simply saying:

“91% Match”

the system explains:

> “This student is a strong match because you specialize in frontend development while this candidate has strong AI/ML and backend skills. Together, the team covers multiple technical areas required for a hackathon project.”

This makes the AI recommendation more understandable and useful.

12. AI TEAM GENERATOR

The application provides an AI Team Generator.

The student can enter:

• Hackathon domain

• Project idea

• Required team size

• Required technologies

The AI generates an ideal team structure.

Example

Project:

AI Healthcare Assistant

Recommended team:

Role	Required Skills
AI/ML Developer	Python, ML, NLP
Backend Developer	FastAPI, Node.js
Frontend Developer	React, JavaScript
UI/UX Designer	Figma, UI Design
Cloud/DevOps	AWS, Docker

13. TEAM SKILL GAP ANALYSIS

After a team is created, the application analyzes the team's skill coverage.

Example:

Skill Area	Coverage
AI/ML	98%
Backend	90%
Frontend	95%
UI/UX	82%
Cloud	65%

The AI identifies:

Skill Gap Detected

> “The team has strong AI and development capabilities but requires additional cloud/deployment expertise.”

The system can then recommend a student with the required skill.

14. PROJECT ANALYSIS

The team can enter its project idea.

Example:

“AI-based crop disease detection system.”

The AI analyzes the project and identifies required skills.

Project Requirements

• Python ✓

• Machine Learning ✓

• Computer Vision ✓

• Frontend ✓

• Backend ✓

• Cloud ⚠

The system can recommend adding a cloud/deployment specialist.

15. TEAM INVITATION SYSTEM

Students can:

• Send invitations

• Accept invitations

• Reject invitations

• Cancel invitations

• Leave teams

Example:

Rahul invited you to join Team Innovators.

Options:

Accept | Decline

16. TEAM DASHBOARD

Once a team is formed, members can view:

• Team members

• Roles

• Skills

• Team compatibility

• Skill coverage

• Skill gaps

• Project idea

• AI recommendations

The dashboard provides a centralized view of the entire team.

17. TECHNOLOGIES USED

Development Platform

Base44 AI

Base44 AI was used to accelerate the development of the web application through AI-assisted application generation and workflow creation.

Application Technologies

The application may include:

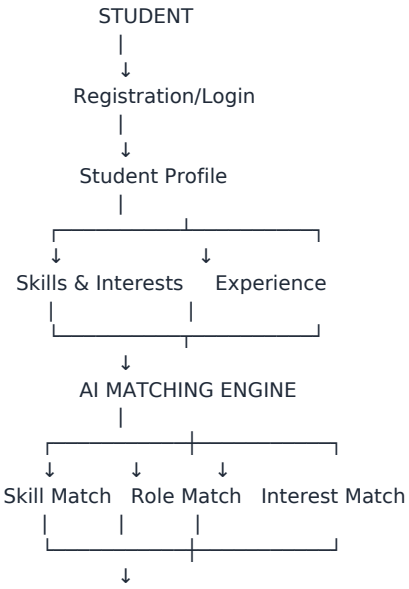
- Frontend UI
- Backend workflows
- Database
- Authentication
- AI-powered matching
- Responsive web design

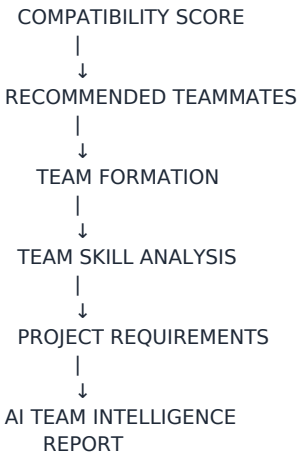
Conceptual Technology Stack

Layer	Technology
UI	Base44 AI
Application Logic	Base44 Workflows
Database	Base44 Database
Authentication	Base44 Authentication
AI	AI-powered matching logic
Deployment	Base44/Web Deployment
Version Control	GitHub

18. SYSTEM ARCHITECTURE

The system can be represented as:





19. DATABASE DESIGN

The application can use the following entities:

Users

Stores authentication and basic user information.

Profiles

Stores student profile information.

Skills

Stores available technical skills.

UserSkills

Connects students with their skills and proficiency levels.

Interests

Stores student interests.

Hackathons

Stores hackathon information.

Teams

Stores team information.

TeamMembers

Connects students with teams.

TeamInvitations

Stores team invitation information.

Matches

Stores AI-generated student matches.

MatchScores

Stores compatibility scores.

Projects

Stores team project information.

Notifications

Stores system notifications.

Messages

Stores team communication.

20. FUNCTIONAL REQUIREMENTS

The system should allow users to:

1. Register and log in.
2. Create and update profiles.
3. Add technical skills.
4. Add interests.
5. Select preferred roles.
6. Search students.
7. Find AI-recommended teammates.
8. View compatibility scores.
9. Understand match explanations.
10. Send team invitations.
11. Create teams.
12. Analyze team skills.
13. Identify skill gaps.
14. Enter project ideas.
15. Analyze project requirements.
16. Generate AI team reports.

21. NON-FUNCTIONAL REQUIREMENTS

Usability

The application should be simple and student-friendly.

Performance

Pages and matching results should load efficiently.

Security

User information should be protected using authentication and authorization.

Scalability

The system should support increasing numbers of students and teams.

Responsiveness

The application should work on desktop, tablet and mobile devices.

Reliability

The application should provide consistent matching and team information.

22. USER INTERFACE

The application follows a modern AI-inspired design.

Main Pages

• Landing Page

• Login

• Registration

• Dashboard

• My Profile

• Find My Team

• Generate Team

• Hackathons

• My Team

• Messages

• Notifications

• Leaderboard

• Settings

Design Theme

• Dark blue

• Electric blue

• Purple

• White

• Gradient backgrounds

• Glassmorphism

• Modern cards

• AI-inspired animations

23. ADVANTAGES

1. Saves time in finding teammates.

2. Provides intelligent recommendations.

3. Focuses on complementary skills.

4. Identifies team skill gaps.

5. Provides explainable AI recommendations.

6. Helps create balanced teams.

7. Improves collaboration.

8. Can be used for different hackathon domains.

9. Easy for students to use.

10. Can be expanded into a larger collaboration platform.

24. LIMITATIONS

1. Matching quality depends on the accuracy of student profiles.

2. AI recommendations may not always perfectly represent human compatibility.

3. Students need to provide correct skill information.

4. Real-time communication features may require additional development.

5. Advanced verification of technical skills requires additional assessment mechanisms.

25. FUTURE ENHANCEMENTS

Future versions can include:

AI Skill Verification

Automatically test whether students actually possess the skills mentioned in their profiles.

GitHub Analysis

Analyze GitHub repositories to estimate technical experience.

Resume Analysis

Upload a resume and automatically extract skills.

Advanced AI Team Formation

Generate multiple possible teams and compare them.

Voice-Based Profile Creation

Students can describe their skills using voice.

AI Project Recommendation

Recommend hackathon project ideas based on the team's combined skills.

Hackathon API Integration

Automatically retrieve upcoming hackathons.

Performance Prediction

Predict the potential strengths and weaknesses of a team.

Gamification

Add badges, achievements and skill levels.

26. TESTING

The application should be tested using different scenarios.

Test Case 1 — Registration

Input: Valid student details

Expected Result: Student account should be created successfully.

Test Case 2 — Profile Creation

Input: Skills, interests and preferred role

Expected Result: Profile should be saved successfully.

Test Case 3 — Team Matching

Input: Student profile

Expected Result: System should display suitable teammates with compatibility scores.

Test Case 4 — Team Invitation

Input: Invite another student

Expected Result: Invitation should be received by the selected student.

Test Case 5 — Skill Gap

Input: Team with no cloud specialist

Expected Result: System should identify cloud as a missing skill.

Test Case 6 — Project Analysis

Input: Hackathon project idea

Expected Result: AI should identify required skills and compare them with team capabilities.

27. EXPECTED OUTPUT

The final application should successfully:

- Create student profiles.
- Analyze student skills.
- Match complementary students.
- Generate compatibility scores.
- Explain AI recommendations.
- Create hackathon teams.
- Analyze team skill coverage.
- Detect missing skills.
- Analyze project requirements.
- Generate team recommendations.

28. SOCIAL AND EDUCATIONAL IMPACT

The project can help students who do not have an existing network of hackathon participants.

It encourages:

• Collaboration

• Peer learning

• Skill development

• Innovation

• Teamwork

• Problem solving

• Technical networking

The platform can be particularly useful in colleges where students from different branches and skill backgrounds need to collaborate.

29. CONCLUSION

The AI Hackathon Team Matcher provides an intelligent solution to the common problem of hackathon team formation.

Instead of relying on friends or random team formation, the system analyzes student skills, interests, roles and experience to recommend complementary teammates.

The project demonstrates how Artificial Intelligence can be applied to a practical student-focused problem.

The use of Base44 AI makes it possible to rapidly develop and prototype the application while focusing on the core innovation: AI-based team formation and explainable teammate recommendations.

The project can be further developed into a complete platform for hackathons, college projects, innovation challenges and collaborative learning.

30. PROJECT LINKS

Live Project

Project Link: https://YOUR-BASE44-PROJECT-LINK

GitHub Repository

GitHub: https://github.com/YOUR-USERNAME/match-hack-syn

Project Name

AI Hackathon Team Matcher

Repository Name

match-hack-syn

31. PROJECT SUMMARY

Category	Details
Project Title	AI Hackathon Team Matcher

Category	Details
Project Type	Mini Project
Domain	Artificial Intelligence / Web Development
Target Users	B.Tech / College Students
Development Platform	Base44 AI
Core Technology	AI-Based Matching
Main Feature	Complementary Skill Matching
Secondary Feature	Team Skill Gap Analysis
AI Feature	Explainable Team Recommendations
Version Control	GitHub
Repository	match-hack-syn
Deployment	Base44
Status	Prototype / Mini Project

32. ONE-LINE PROJECT DESCRIPTION

“An AI-powered web application that intelligently matches students based on complementary skills, interests and roles to create balanced hackathon teams.”

33. GITHUB README SHORT DESCRIPTION

AI Hackathon Team Matcher is an AI-powered web application that helps students find suitable hackathon teammates based on their skills, interests, experience and complementary abilities. The platform provides compatibility scores, explainable recommendations, team formation, skill-gap analysis and AI-powered team insights.

Built using: Base44 AI

Repository: match-hack-syn